

Chenyi Xu

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Education

Hangzhou Dianzi University

B.Sc. in Computer Science and Technology GPA: 3.65 / 4.0

Core Courses: Advanced Mathematics, Probability Theory, Linear Algebra, Software Engineering, Computer Networks

Hangzhou, China

2023.09 – 2027.06

Research & Projects

Embodied AI Research

Research Assistant, School of Computer Science, Hangzhou Dianzi University

Hangzhou, China

2025.5 – 2026.6

- Investigated Vision-Language-Action (VLA) models for robotic manipulation, optimizing end-to-end policy learning with latent action representations and spatial grounding to improve generalization across unseen objects and environments.
- Explored Vision-and-Language Navigation (VLN) in photorealistic 3D environments, developing history-aware visual-text cross-modal matching and adaptive waypoint planning to reduce navigation drift in long-horizon tasks.

4D Vision Reconstruction

Visiting Student, AGI Lab, Westlake University

Hangzhou, China

2025.07 – 2025.09

- Proposed a monocular dynamic video 4D reconstruction framework based on 4DGS, using single-step diffusion models and information-gain-driven pose selection to generate high-quality pseudo-views for multi-view supervision.
- Modeled scene dynamics using a canonical-space 4DGS representation and a globally shared $SE(3)$ motion basis trajectory, integrating a Neural Texture Field to achieve stable long-sequence reconstruction.

JEPA-based VLA Learning

Research Intern, HAI Lab, Eastern Institute of Technology

Ningbo, China

2026.07 – Present

- Developed a JEPA-based Vision-Language-Action framework for embodied intelligence, leveraging predictive latent representations to learn task-relevant visual dynamics from large-scale video and robot interaction data.
- Modeled future state transitions in latent embedding space with self-supervised video prediction objectives, integrating learned representations into VLA policies to improve generalization and data efficiency in robotic manipulation tasks.

Publications

L. Yan[†], Y. Wu, **C. Xu**, C. Yang, J. Zhang, & P. Li. [AR-Nav Benchmark: Augmented Reality Navigation with Vision and Language](#). *AAAI Conference on Artificial Intelligence* ([†]Mentor). (**AAAI 2026, Oral, CCF-A**).

- Proposed the AR-VLM model to construct high-fidelity 3D semantic fields via symbolic geometric alignment, resolving the long-term memory deficiency in AR navigation.
- Developed the ARN-Pilot module for lightweight topological reasoning and sub-second path planning, establishing the first standardized multi-scenario AR-Nav benchmark framework.

M. Yang, Y. Yang, **C. Xu**, C. Song, Y. Zuo, T. Zhao, R. Li, & C. Zhang[†]. [Fast3Dcache: Training-free 3D Geometry Synthesis Acceleration](#). *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR 2026, CCF-A**).

- Proposed Fast3Dcache, a training-free 3D geometric caching framework leveraging voxel stabilization regularities during 3D diffusion generation to accelerate inference.
- Designed dynamic cache scheduling (PCSC) and stability discrimination (SSC) based on motion features, achieving a 27% speedup and 54% FLOPs reduction while preserving geometric quality.

C. Xu, Y. Wu, L. Yan[†], C. Yang, J. Zhang, F. Guan, & P. Li. [LongDPM: Overlap-Aware 4D Reconstruction from Long Monocular Videos](#). In *Chinese Conference on Pattern Recognition and Computer Vision* (**PRCV 2026, CCF-C**).

- Proposed the LongDPM framework for long video 4D reconstruction via overlap-aware chunked inference, separating static anchors from dynamic features to ensure robust cross-chunk registration.
- Developed a geometrically-conditioned trajectory association and fusion mechanism to maintain identity continuity across chunks, achieving SOTA performance across multiple benchmarks.

Y. Wu, C. Cai, L. Yan[†], M. Li, **C. Xu**, C. Yang, F. Guan, J. Zhang, & P. Li. [ShapeNav: 3D Geometric Perception and Reasoning for Multi-Floor Vision-and-Language Navigation](#). *ACM International Conference on Multimedia* (**ACM MM 2026, Oral, CCF-A**).

- Proposed ShapeNav, a 3D geometric navigation framework without 3D pre-training. It integrates point clouds and vision-language models, with GPE and MOT modules to align 3D geometric and 2D visual features for cross-modal fusion.
- Achieved SOTA navigation results on the R2R and F2F benchmarks, with a 3% higher success rate and a 0.29m lower navigation error than DUET. The framework retains strong performance even when only half of the training data is available.

C. Xu*, Y. Wu*, L. Yan[†], Z. Bai, Q. Wang, F. Guan, J. Zhang, & P. Li. [Dense Soccer Video Captioning with Global Context and Player Position Awareness](#). (**Scientific Reports, JCR-Q1**).

- Proposed the SCPR framework for dense soccer video captioning, leveraging Perceiver attention (GCPR/PCPR modules) to fuse global-local video context with multi-modal player positions.

- Integrated an SAA spatial enhancement module to inject pitch tactical statistics for LLM decoding, yielding optimal fine-grained event localization and SOTA performance on public benchmarks.
- Y. Wu*, C. Xu*, L. Yan†, C. Cai, G. Min, F. Guan, B. Lin, J. Zhang, & P. Li. [BrainNav: Towards Dual-Brain Minimal Sufficient Representation for Vision-Language Navigation](#). (*Equal contribution). (Submitted to ICRA 2027, CCF-B).
- Proposed the BrainNav dual-brain framework for long-horizon navigation, utilizing instruction-driven attention filtering to eliminate irrelevant visual noise and enhance language-spatial alignment.
 - Constructed a low-rank constrained alignment and a compact world model for state prediction within compressed latent spaces, achieving SOTA performance on R2R and RxR benchmarks.
- C. Yang*, C. Xu*, L. Yan†, Z. Li, G. Min, F. Guan, J. Zhang, & P. Li. ToolUse: A Generative Robotic Benchmark for Language-Guided Open-World Tool Manipulation. (Submitted to ICASSP 2027, CCF-B).
- Proposed Tool-VLA, a geometry-aware vision-language-action model for open-world robotic tool manipulation. It leverages semantic-geometric alignment (SGA) to extract tool functional frames, augmenting language prompts with grasp semantics, functional-part descriptions, and execution directions for geometry-grounded cross-modal policy learning.
 - Established the ToolUse generative benchmark covering multi-tier tool-use tasks. Experiments show that simulation pre-training on ToolUse brings over 40% real-world success improvement compared with real-data-only training and achieves consistent gains under limited training-data proportions.

Awards, Honors & Competitions

CUMCM “Higher Education Press Cup” Provincial First Prize, National Third Prize	2024.09
Service Outsourcing Innovation & Entrepreneurship Competition National Third Prize	2024.12 – 2025.03
ICT Industry–Education Integration Innovation Competition National Second Prize	2024.08 – 2024.10
ICT Industry–Education Integration Innovation Competition National Second Prize	2025.08 – 2025.10
Zhejiang Provincial Xinmiao Talent Program Project Leader	2025.12 – 2026.12
First-Class Academic Scholarship & Outstanding Student Leader Award HDU	2024 & 2025

Skills & Leadership

- **Campus Leadership:** Class Monitor (Class 23052314) — Class Assistant (Class 25050417).
- **Languages:** English (CET-4 & CET-6 qualified).
- **Intellectual Property:** 7 Invention Patents filed/authorized (Public Nos.: CN122597679A, CN122524119A, CN122392071A, CN122322854A, CN122518363A, CN122378690A, CN120744017A).